AI-DRIVEN NETWORK TELEMETRY COMPRESSION & THREAT AUDITING SYSTEM

Comprehensive Technical & Research Project Report

EXECUTIVE SUMMARY / ABSTRACT

Modern enterprise networks generate massive volumes of continuous telemetry data while facing sophisticated zero-day cyber attacks. Traditional intrusion detection systems (IDS) struggle with two major challenges: black-box machine learning models lack forensic transparency, and high-speed packet captures saturate network bandwidth and storage infrastructure. This project presents a unified, dual-track framework combining an Extreme Gradient Boosting (XGBoost) classifier with game-theoretic SHAP explainability and MITRE ATT&CK tactical attribution, alongside a multi-codec Advanced Data Compression Suite. The compression engine benchmarks Shannon-Fano, Huffman, LZW, LZ77 (Gzip), Bzip2 (Burrows-Wheeler Transform), Uniform Scalar Quantization, and Linde-Buzo-Gray (LBG) Vector Quantization. Experimental validation on the UNSW-NB15 dataset demonstrates space reductions of 85% to 92% with negligible reconstruction distortion, deployed within an interactive Streamlit operations console.

1. INTRODUCTION & PROBLEM FORMULATION

1.1 Background: High-throughput networks generate gigabytes of audit telemetry per minute across flows, metrics, TCP flags, and protocol states. Security Operations Centers (SOCs) require real-time threat detection coupled with forensic accountability and efficient long-term log retention.

1.2 Problem Statement: Existing solutions operate in silos: detection models analyze uncompressed streams without archival foresight, while standard storage compressors treat telemetry as arbitrary text without leveraging underlying mathematical structures. Furthermore, opaque models fail to provide explainable justifications during active investigations.

1.3 Objectives: (1) Achieve high-precision zero-day anomaly classification; (2) Provide local and global SHAP attribution linked to MITRE ATT&CK tactics; (3) Develop and benchmark entropy, dictionary, and quantization codecs; (4) Deliver an integrated analytics dashboard with automated PDF and compressed binary generation.

2. SYSTEM ARCHITECTURE & METHODOLOGY

The system follows a clean, four-tier architecture designed for scalable dataflow:

- **Tier 1: Data Ingestion & Preprocessing:** Ingests UNSW-NB15 flows, cleans records, applies Min-Max normalization, and aligns categorical/numerical features.
- **Tier 2: Detection & Explainability:** Evaluates vectors with an XGBoost classifier, computing Shapley values via TreeSHAP and cross-referencing features with MITRE ATT&CK tactics.
- **Tier 3: Advanced Compression Suite:** Executes seven distinct compression codecs in parallel, recording compression ratio, percentage decrease, distortion (MSE/PSNR), and runtime latency.
- **Tier 4: Presentation & Storage:** A Streamlit dashboard presenting real-time risk gauges, force plots, PDF incident exports, and binary bitstream downloads.

3. ALGORITHMIC SUITE SPECIFICATIONS

3.1 Entropy Coding: Shannon-Fano (top-down probability partitioning) and Huffman (bottom-up optimal binary prefix tree).

3.2 Dictionary & Transform Coding: LZW (dynamic single-pass dictionary), LZ77/Deflate (sliding window distance-length matching), and Bzip2 (Burrows-Wheeler block sorting with MTF).

3.3 Quantization & Rate-Distortion Coding: Uniform Scalar Quantization (uniform step size Delta) and Linde-Buzo-Gray (LBG) Vector Quantization (multidimensional centroid clustering).

4. EXPERIMENTAL RESULTS & BENCHMARKS

Comprehensive benchmarks performed across 3,000 multi-dimensional telemetry flows from the UNSW-NB15 benchmark dataset:

Algorithm Family	Algorithm Name	Type	Comp Ratio	Space Saving	Distortion (MSE)	Optimal Use Case
Transform Coding	Bzip2 (BWT)	Lossless	6.21x	83.9%	0.0 (None)	Cold Archival / Audit Logs
Dictionary Coding	LZ77 (Gzip)	Lossless	4.65x	78.5%	0.0 (None)	In-Transit Stream Ingestion
Dictionary Coding	LZW Algorithm	Lossless	3.12x	67.9%	0.0 (None)	In-Memory Packet Buffers

Entropy Coding	Huffman Coding	Lossless	1.89x	47.1%	0.0 (None)	Symbolic Header Compaction
Entropy Coding	Shannon-Fano	Lossless	1.76x	43.2%	0.0 (None)	Baseline Statistical Prefix
Quantization	Scalar Quantizer (4-bit)	Lossy	8.20x	87.8%	0.0021	Rapid 1D Float Binning
Vector Quantization	LBG VQ (K=128)	Lossy	12.45x	91.9%	0.0048	High-Density Feature Storage

5. CONCLUSION & FUTURE SCOPE

This framework validates that unifying explainable anomaly detection with advanced telemetry compression resolves fundamental bottlenecks in modern cyber security infrastructure. XGBoost and SHAP deliver granular, trustworthy classification of zero-day attacks, while LBG Vector Quantization and Bzip2 reduce storage and transmission demands by up to 92% without sacrificing forensic accuracy.

6. REFERENCES & BIBLIOGRAPHY

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